



Improved compositional analysis of block copolymers using Diffusion Ordered NMR Spectroscopy

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ABSTRACT

Block copolymers constitute a fascinating class of polymeric materials that are used in a broad range of applications. The performance of these materials is highly coupled to the physical and chemical properties of the constituting block copolymers. Traditionally, the composition of block copolymers is obtained by ¹H NMR spectroscopy on purified copolymer fractions. Specifically, the integrals of a properly selected set of ¹H resonances are compared and used to infer the number average molecular weight (M_n) of one of the block from the (typically known) M_n value of the other. As a corollary, compositional determinations achieved on imperfectly purified samples lead to serious errors, especially when isolation of the block copolymer from the initial macro initiator is tedious. This investigation shows that Diffusion Ordered NMR Spectroscopy (DOSY) can be used to provide a way to assess the advancement degree of the copolymerization purification/reaction, in order to optimize it and hence contribute to an improved compositional analysis of the resulting copolymer. To this purpose, a series of amphiphilic polystyrene-*b*-poly(ethylene oxide) block copolymers, obtained by controlled free-radical nitroxide mediated polymerization, were analyzed and it is shown that, under proper experimental conditions, DOSY allows for an improved compositional analysis of these block copolymers.

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1. Introduction

Block copolymers constitute a fascinating class of polymeric materials made by the covalent bonding of at least two macromolecular chains. This gives rise both in solution and in the bulk to a rich variety of microstructures which are used in a broad range of applications from advanced nanomaterials to biocompatible drug delivery systems [1–3]. The performance of these materials is highly coupled to the physical and chemical properties of the constituting block copolymers, especially on their chemical composition and molecular weight distribution [2]. Typically, block copolymers are synthesized through an extensive array of polymerization techniques [1]. However, due to unavoidable side reactions in these methodologies, copolymer samples are most often very complex systems characterized by a mixture of various polymeric chains. These include residual and/or side-produced homopolymers in addition to the block copolymer itself, all these macromolecules presenting compositional and molecular weight heterogeneity which sometimes leads to poor physicochemical

properties. Purification procedures are thus required to isolate the desired block copolymer prior to its characterization. To this purpose, the batch fractionation method was traditionally employed. By capitalizing on the distinct miscibility of the various macromolecules, the desired copolymer is isolated through selective precipitation. While being efficient in most cases, this method remains empirical and is not necessarily straightforward to optimize. While the method cumulative efficiency can be enhanced by performing several batch fractionations successively, isolation of the block copolymer sample may still be incomplete.

This problem may be somewhat avoided by the development of preparative scale Size Exclusion Chromatography (SEC) methods. Here the separation between the different macromolecular entities is based on differences in their respective hydrodynamic size. However, these methods require a high level of expertise because uncontrolled polymer/support interactions must be totally avoided. Moreover it is well established that, for a given block copolymer, variations in the relative size of the blocks may slightly modify the experimental set up and hence compromise the separation efficiency [4,5]. Finally, as a time- and solvent-consuming technique, SEC may not be desirable when rapid analysis is required. This basically implies that SEC fractionation is not systematically performed on a routine basis. As a matter of fact proofs are rarely provided

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in the literature to confirm that the batch fractionation procedure has been totally successful and that the desired copolymer is actually the only macromolecule present in solution, even though its presence is systematically demonstrated. More recently, new liquid chromatography techniques such as liquid adsorption chromatography (LAC) or liquid chromatography at the critical adsorption point [4,5] (LC-CAP), were developed to determine the chemical composition of block copolymers. However, these techniques are still limited because of the difficulty in finding the right set of experimental conditions, which are usually specific to one block copolymers family.

Compositional studies of polymerized block copolymers are traditionally performed by ^1H NMR. More precisely, the integrals of a properly selected set of ^1H resonances are compared and used to infer the number average molecular weight (M_n) of one of the block from the (typically known) M_n value of the other. Provided that the ^1H spectrum has been recorded using quantitative experimental parameters (i.e., enough time has been allowed for the resonances to relax completely), and no spectral overlapping is present, the estimation is rather accurate. However, because it is essentially impossible to differentiate the NMR signals due to a given homopolymer from those due to the corresponding block of the block copolymer, an incomplete fractionation will lead to miscalibrated integrals and hence to meaningless M_n values.

In this context, we propose the DOSY (Diffusion Ordered Spectroscopy) NMR experiment as a fast and reliable technique to identify and account for the presence of residual homopolymers in solution. Based on the Pulsed Gradient Spin Echo (PGSE) experiment [6–8], DOSY is a very convenient processing scheme which produces two-dimensional correlation maps showing chemical shifts and diffusion coefficients on the horizontal and vertical axes, respectively [9]. Many applications of DOSY have been reported in the literature [10], some of which being directly related to the characterization of homopolymers [11–16] or graft copolymers [11,17]. For instance, Zawaneh et al. have used DOSY to evidence that their synthesized block copolymers had not been degraded during a specific deprotection reaction [18]. Interestingly, however, while a few investigations have already reported the use of DOSY for investigating block copolymer supramolecular structures [19], no study has clearly addressed so far, to the best of our knowledge, the potential of this technique for the simple compositional analysis of block copolymers.

2. Experimental

2.1. Materials

All chemicals were purchased from Sigma–Aldrich and were used without further purification. Deuterated solvents (CDCl_3) were provided by Euriso-Top. Details on the synthesis of the PS-*b*-PEO block copolymers used in this study can be found in Ref. [20].

2.2. DOSY experiments

DOSY experiments were performed on a Bruker Avance spectrometer operating at 500 MHz for the ^1H Larmor frequency with a 5-mm triple resonance inverse Bruker cryoprobe optimized for ^1H detection and equipped with an actively shielded z-gradient coil. The gradient coil was calibrated by measuring the self-diffusion coefficient of the residual proton in D_2O [21] and was found to be 55 G cm^{-1} . The temperature was set and controlled to 300 K with an air flow of 545 L h^{-1} to avoid temperature fluctuations due to sample heating during the gradient pulses. CDCl_3 solutions (3 mg mL^{-1}) were prepared by weighing an amount of sample directly into

the NMR sample tube and adding 0.6 mL of deuterated solvent. The pulse sequence used for the measurements was based on a double-stimulated echo and incorporated bipolar gradients and a longitudinal eddy current delay, used for minimizing spectral artefacts resulting from eddy currents and avoiding complications due to thermal convection [22]. In this pulse sequence, the amplitude of a NMR resonance observed at the echo I is given by [23]:

$$I = I_0 \exp(-D(\gamma g \delta)^2 (\Delta - \varepsilon(\delta)))$$

where I_0 is the resonance amplitude at zero gradient strength, γ is the magnetogyric ratio of the observed nucleus, g and δ are the strength and the duration of the gradient pulses, respectively, and Δ is the diffusion time, i.e., the time during which the diffusion is monitored. $\varepsilon(\delta)$ is a correction factor that depends on both δ and the pulse sequence. Usually, all delays are kept constant to avoid any complication arising from magnetic relaxation, and only the gradient strength is varied. Specifically, the gradient strength was quadratically incremented in 16 steps from 6 to 95% of its maximum value. The diffusion time and the duration of the bipolar sine gradient pulses were optimized for each experiment to achieve at the largest gradient amplitude a decrease in the resonance intensity higher than 95%. The gradient pulse recovery time and the longitudinal eddy current delay were set to 0.25 and 25 ms, respectively. After Fourier transformation and phase correction, the baseline of the spectra was carefully adjusted. The diffusion dimension of the 2D DOSY spectra was processed by means of the Bruker XwinNMR™ software package (version 3.5). The DOSY maps were obtained with the XwinNMR software (version 3.5) by selecting a monoexponential function and choosing 128 points in the vertical dimension. The plots shown in Fig. 1 are contour plots optimized to show the copolymer signals. Other impurities, such as the residual solvent and water signals, were also detected in the spectra but do not appear necessarily on the spectra because of the threshold used for building the contour plots. Note that, in this kind of representation, the width of the correlation spots in the vertical dimension does not exclusively depend on the fitting precision but also on internal software parameters. This width also depends on the threshold used for building the 2D contour plot. Indeed, DOSY maps are typically constructed to maintain the relative signal

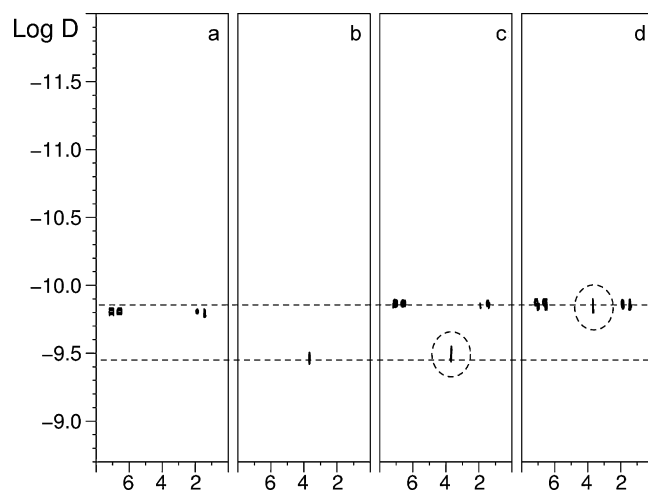


Fig. 1. DOSY maps recorded in CDCl_3 at 300 K on a 500 MHz spectrometer for 3 distinct polymer samples (a) PS homopolymer, $M_w \sim 10\text{ kDa}$; (b) end-functionalized PEO homopolymer, $M_w \sim 2\text{ kDa}$; (c and d) block PS-*b*-PEO copolymer obtained by NMP using the end-functionalized PEO sample reported in (b) after an incomplete (c) and complete (d) batch fractionation (selective precipitation in ethanol). The upper and lower dashed horizontal lines shown in the figure indicate the D values of the PS-*b*-PEO copolymer and the end-functionalized PEO homopolymer samples, respectively, whereas the dashed circles are only used to evidence the D value characterizing the PEO ^1H NMR signal.

intensity ratio that is observed in the corresponding ^1H spectrum (provided of course that the mixture components do not display dramatic difference in relaxation times). In other words, an intense signal in the ^1H spectrum will give rise to an intense cross-peak in the DOSY map. Because the ^1H signal due to the PEO block (singlet) is usually more intense than those due to the PS block (multiplets), at least in this work when considering the relative average molecular weight of the blocks of the analyzed copolymers, the cross-peak characterizing the PEO block will be more intense than those characterizing the PS block in the DOSY map. Clearly, to show both types of cross-peaks, the threshold value that is selected to build the 2D contour plot must necessarily be low. As a result, the difference between the threshold value and the signal intensity will be much larger for PEO than for PS. Overall, this will translate into an apparent larger signal width for PEO, as illustrated in Fig. 1c and d. Finally, the data were also analyzed by plotting the signal intensities (areas) as a function of the gradient strength and fitting the resulting decays to the previously reported equation using a nonlinear least-squares fit based on the Levenberg–Marquardt algorithm. The experimental precision of the diffusion coefficients reported in this investigation was $\pm 4\%$, which resulted from a compromise between sensitivity (dilute solutions were considered) and total experimental time (~ 30 min). Higher precision could be achieved at the expense of longer acquisition times.

3. Principle

In the basic implementation of DOSY, ^1H spectra with increasing diffusional attenuation are collected and diffusion coefficients for the ^1H resonances are subsequently calculated by fitting the exponentially decaying signals to the appropriate expression [8]. One approach consists of constructing DOSY maps by fitting the data with a monoexponential function. In this way, no user biases are introduced in the data analysis and hence spurious results are avoided [10]. Obviously, the accuracy of this approximation relies on the assumption that the ^1H resonances due to species of distinct molecular weight, and hence of distinct diffusion coefficients, do not overlap. This condition is likely violated in the case of mixtures of homo- and co-polymers carrying the same block. Here, the use of a monoexponential function necessarily results in the identification of a single (and thus meaningless) apparent diffusion coefficient, roughly equal to the weighted average of the diffusion coefficients characterizing the various species. However, such simplified processing can still be useful for monitoring the degree of advancement of copolymerization reactions. In the case of a successful block copolymerization through, for instance, the macromolecular coupling of two well-characterized X and Y homopolymers, the correlation spots characterizing the ^1H NMR signals due to the X and Y blocks of the X-*b*-Y copolymer must appear on the same trace on the diffusion dimension. In other words, all the resonances in the ^1H spectrum must be characterized by a same diffusion coefficient which, because of the higher molecular weight of the resulting X-*b*-Y block copolymer, must be lower than the diffusion coefficient of the X and Y homopolymers. Any deviation from this limit case, such as the presence in solution of both the X homopolymer and the X-*b*-Y copolymer, will result in a shift of the apparent diffusion coefficient associated with the X resonances towards the value specific to the homopolymer. Therefore, monoexponential DOSY processing definitely appears as an adequate approximation because any deviation from pure monoexponential behaviour already indicates that the copolymerization procedure has not been optimal. Similar observations hold for copolymers synthesized using a macroinitiator through other polymerization reactions, e.g. controlled free-radical polymerization. Clearly, the main limitation of this approach lies in the precision

with which the diffusion coefficients are measured. The precision of high resolution DOSY in the absence of spectral overlapping has been claimed to be as good as 1% [24], the only limiting factor being the sensitivity as expressed by the signal-to-noise ratio of the NMR spectrum. This means that, even in favourable cases (high signal-to-noise ratio), there will always be a limiting Y block size of the X-*b*-Y diblock copolymer in order for the copolymer to be distinguished from the residual X homopolymer (roughly estimated to be between 2 and 4%).

4. Results and discussion

As a proof of principle, we selected a series of polystyrene/poly(ethylene oxide) amphiphilic block copolymer (PS-*b*-PEO) samples which are industrially significant because of their remarkable biological and chemical properties. These polymers can form highly ordered, self-assembled nanostructures that have been recently used as mesoporous silica films of different pore sizes [20] and electrolytes for rechargeable batteries [25]. Specifically the block PS-*b*-PEO samples under study were obtained by controlled free-radical nitroxide mediated polymerization (NMP) [26]. Indeed, NMP is an efficient technique that has recently attracted increasing interest because it allows the polymerization to occur in a living fashion with a high degree of control over the entire polymerization process [20]. In our case, the macroinitiator was obtained as described in Ref. [27] by functionalizing a well-characterized, monodisperse (polydispersity index $I_p < 1.1$) PEO homopolymer sample. All PS-*b*-PEO copolymers samples were batch fractionated by using selective precipitation with ethanol. DOSY experiments were then performed on dilute CDCl_3 solutions of these samples. Specifically, in a DOSY map, the NMR signals of a given set of nuclei are correlated with the diffusion coefficient of the molecule to which they belong, provided that no intermolecular interactions [28] or chemical exchange are present [29,30]. The data inversion step undoubtedly represents the key to DOSY, as it is typically impossible to guess *a priori* the number of components to fit when analyzing unknown mixtures. Moreover, to simulate the result of an unsuccessful or incomplete fractionation, a small amount of the end-functionalized PEO homopolymer was added to the copolymer solutions, and the resulting samples were also analyzed by DOSY. As an illustration, results obtained for one of these PS-*b*-PEO samples are reported in Fig. 1. Note that the NMP was stopped to reach a PS block characterized by an average molecular weight of about 10 kDa. Therefore, a dilute CDCl_3 solution of a monodisperse PS homopolymer (M_n 10 kDa) was also analyzed by DOSY for comparison. Fig. 1a and b show the results of the DOSY experiments recorded on dilute CDCl_3 solutions of the PS and the end-functionalized PEO homopolymer samples, respectively. The ^1H signals due to the PS sample, resonating both in the 7.2–6.3 ppm and 2.0–1.3 ppm spectral regions, are all aligned on the same horizontal line. Similarly in Fig. 1b the ^1H signal due to the PEO sample (3.6 ppm) is characterized by a larger diffusion coefficient. In parallel, Fig. 1c and d show the results obtained on dilute CDCl_3 solutions of the PS-*b*-PEO block copolymer before (Fig. 1c) and after (Fig. 1d) complete batch fractionation. Clearly, Fig. 1d shows that the ^1H PEO signal (evidenced by the dashed circle) is perfectly aligned on the horizontal line characterizing the ^1H PS signals. This implies that all these ^1H signals are due to the same macromolecule whose diffusion coefficient is lower than that of the PS and PEO homopolymers. Therefore DOSY indicates that the copolymerization reaction was carried out successfully and that only the PS-*b*-PEO copolymer is present in solution. In this case, because the above mentioned ^1H signals are strictly due to the PS and PEO blocks of the PS-*b*-PEO copolymer, careful integration of these signals in the corresponding ^1H spectrum can accurately reveal the M_n value of the PS

Table 1

Diffusion coefficients (given with a $\pm 4\%$ precision as described in the Section 2) of the samples considered in this study, as measured by DOSY experiments performed in CDCl_3 solutions at 300 K.

Sample		$D (\times 10^{-10} \text{ m}^2 \text{ s}^{-1})^a$	R^a
Homo PS (~ 10 kDa)		1.52	0.9997
Homo PEO (~ 2 kDa)		3.32	0.9999
PS- <i>b</i> -PEO (pure sample)	PS block signal	1.31	0.9992
	PEO block signal	1.31	0.9990
PS- <i>b</i> -PEO (in a mixture with the PEO initiator)	PS block signal	1.31	0.9995
	PEO block signal ^b	2.61	0.9984
	PEO block signal ^c	3.21 (80%); 1.32 (20%)	0.9999

^a Adjusted *R*-square value of the fit.

^b Monoexponential fit.

^c Biexponential fit.

block by using the known M_n value of the end-functionalized PEO homopolymer.

On the contrary, Fig. 1c shows that the PS and PEO ^1H signals are not characterized by the same diffusion coefficient. Performing the compositional analysis on a sample characterized by such a DOSY map would clearly lead to an incorrect composition (the M_n value of the PS block will be overestimated). A finer analysis of the PEO signal decay in the PGSE experiment using a biexponential function allowed us to obtain for this signal two distinct diffusion coefficients: $1.32 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$ and $3.21 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$ (Table 1). Within our experimental error ($\pm 4\%$), the former corresponds to that of the PS-*b*-PEO copolymer, which presence in solution was hence confirmed, whereas the latter corresponds to that of the unreacted end-functionalized PEO homopolymer. In other words, the position of the PEO correlation spot in the simplified DOSY processing showed in Fig. 1c echoes the presence of both species. In principle the amounts of the two components obtained from the biexponential fitting would allow the relative concentration of the two species to be estimated. Indeed, these proportions could be used to recalibrate the integral of the PEO signal (used as a reference) in the corresponding quantitatively recorded ^1H NMR spectrum. However, because of possible complications arising from differences in the magnetic relaxation properties of the involved polymers, care should be taken when applying such method, which should only be regarded as a way to achieve a semi-quantitative estimation. Nonetheless the progress of the purification/reaction can already be fairly evaluated from the position of the PEO spot, which is expected to be closer to that of the PS peaks as the relative copolymer concentration increases.

5. Conclusion

DOSY has been proposed as a fast (10–30 min of experimental time required) and efficient way to monitor the advancement of purification procedures, hereby improving the compositional analysis of block copolymers. Although the separation mechanism involved in DOSY and SEC is essentially the same, in a sense that both techniques distinguish the polymers according to difference in their respective hydrodynamic size, the absence of a stationary phase is a clear advantage of DOSY over SEC as no adverse adsorption effect would be deployed. However, similarly to SEC, the DOSY methodology proposed here should fail whenever the hydrodynamic size of the block copolymer is comparable to that of the homopolymer used as the macroinitiator, since the difference in size between the residual homopolymer and the block copolymer would be difficult to reveal. However this situation is rather unlikely when considering the typical block sizes in use, and hence high quality NMR diffusion data should allow even small polymer size changes to be detected (roughly estimated to lie at best between 2 and 4% in case of an optimal sensitivity). As a further word of

caution, because the mathematical inversion of the PGSE signal decays is an ill-conditioned problem, complications may arise in the DOSY analysis when NMR signals overlap, especially for similar molecular weight polymers or for polydisperse samples. In these unfavourable cases, the method presented here may at first sight lead to very cumbersome or even intractable calculations. However, many strategies have been developed to overcome these difficulties [11,31–35], and future progresses in this area are expected to turn DOSY into a very general analytical tool for polymer characterization. Finally, because diffusion data can be used to estimate the weight average molecular weight (M_w) of homopolymers, provided that specific experimental precautions are taken [14,16], it can already be anticipated that, for block copolymers, DOSY could be of use for the characterization of the blocks M_w values, and investigations along these lines are currently in progress.

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